Learning Chaotic Dynamics on the Lorenz System

Scientific Machine Learning Course Project, Yimng Liu

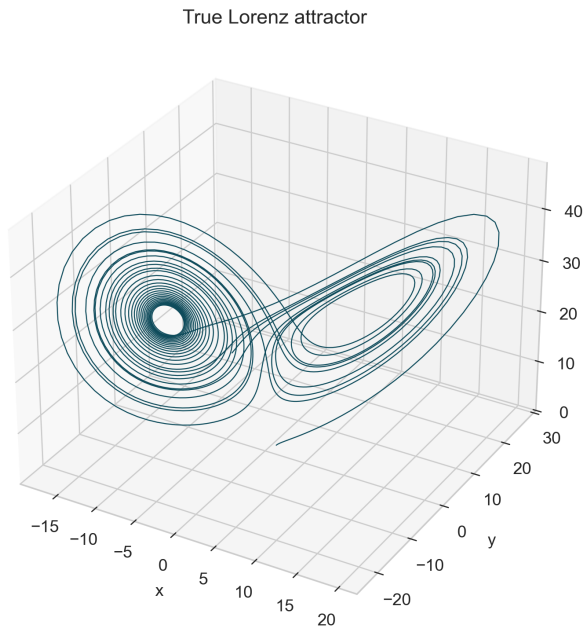

Fig. 1. Reference Lorenz attractor.

***Abstract*—This report studies learned forecasting and dynamical reconstruction on the chaotic Lorenz system using three model families: an MLP, a ResNet, and a Neural ODE. The ResNet achieves the best one-step mean-squared error, while the Neural ODE achieves the best rollout RMSE at horizons 50, 100, and 500 and shows the strongest robustness under perturbation. The main result is that chaotic systems should not be judged by local prediction error alone: rollout stability, attractor reconstruction, and robustness are also necessary to evaluate whether a learned model captures meaningful dynamics.**

I. Introduction

Scientific Machine Learning (SciML) combines numerical modeling and machine learning for systems governed by scientific structure. The Lorenz system provides a compact and classical benchmark for this purpose:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z, \tag{1}$$

with $(\sigma, \rho, \beta) = (10, 28, 8/3)$ under the standard chaotic setting. Writing $s(t) = [x(t), y(t), z(t)]^\top$, the system can be expressed as $\dot{s}(t) = f(s(t))$.

The Lorenz equations are deterministic but chaotic, with strong sensitivity to initial conditions and the well-known butterfly-shaped strange attractor shown in Fig. 1. This makes the system a useful testbed for learned dynamics. A model may achieve very low one-step prediction error while still failing to generate stable long-horizon trajectories. Therefore, the main question of this project is not only whether a model predicts the next state accurately, but also whether it preserves meaningful long-term dynamical structure.

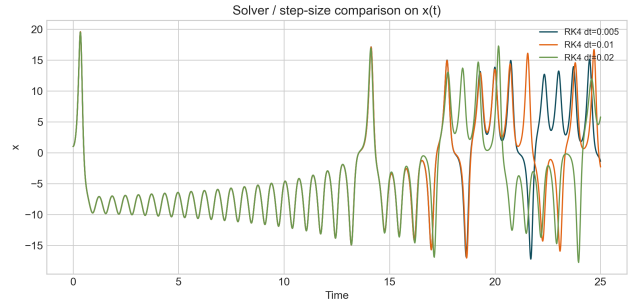

Fig. 2. RK4 step-size comparison on $x(t)$.

Motivated by this challenge, the project is organized around three evaluation goals: short-horizon forecasting, rollout stability, and attractor reconstruction. For chaotic systems, pointwise trajectory agreement becomes fragile over time, so local accuracy must be distinguished from qualitative dynamical fidelity.

II. Method

Reference trajectories were generated by fixed-step fourth-order Runge–Kutta integration and split by trajectory into 100 training, 20 validation, and 20 test trajectories. Each trajectory was simulated over $T = 25$ with default step size $\Delta t = 0.01$.

We compare three models. The MLP directly predicts s_{t+1} and serves as the simplest baseline. The ResNet predicts an increment and uses

$$s_{t+1} \approx s_t + \Delta s_t, \tag{2}$$

which is consistent with the first-order numerical structure $s_{t+\Delta t} \approx s_t + \Delta t\, f(s_t)$. The Neural ODE instead learns a continuous vector field f_θ and obtains trajectories by integrating $\dot{s} = f_\theta(s)$, making it the model most closely matched to the original continuous-time formulation.

The evaluation follows the same logic as the presentation. First, short-horizon forecasting is measured by one-step MSE. Second, long-horizon behavior is measured by rollout RMSE at horizons 50, 100, and 500. Third, global phase-space fidelity is measured by attractor-level similarity, quantified through an attractor-density distance, where lower values indicate better reconstruction. Additional robustness experiments consider step-size variation and observation noise. Figure 2 further shows that even the same ODE produces visibly different long-time trajectories when the numerical step size changes, highlighting the intrinsic difficulty of exact long-horizon matching in chaotic systems.

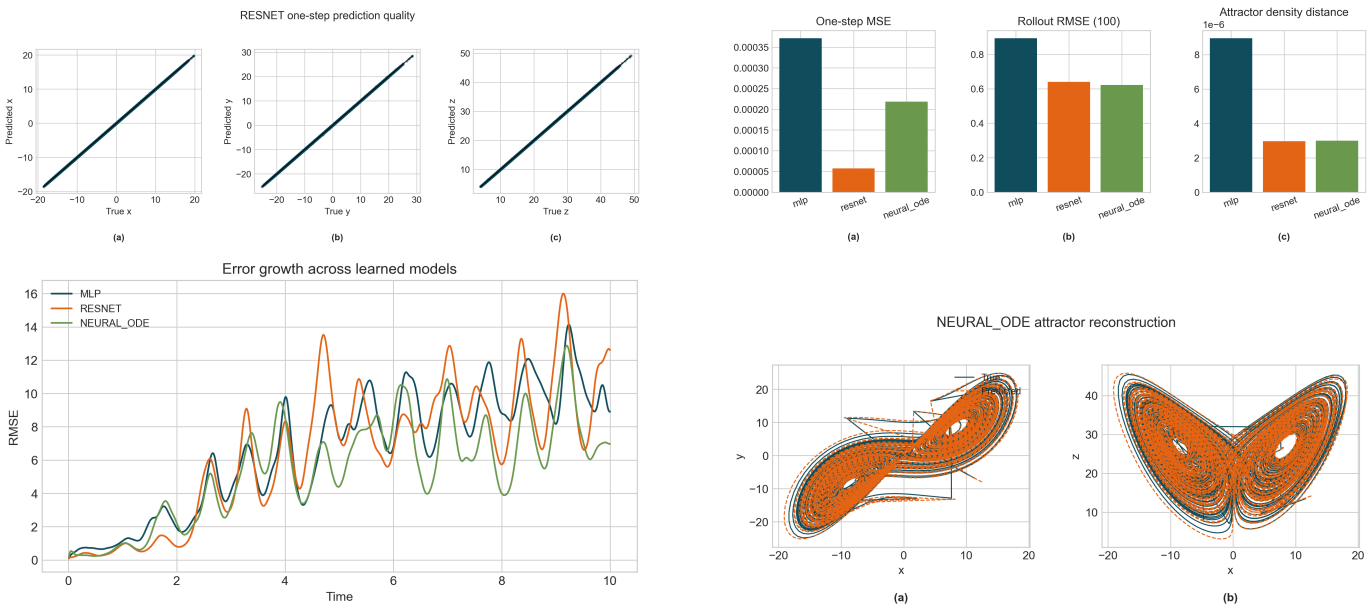

Fig. 3. Top-left: ResNet one-step prediction quality. Top-right: compact overall metric summary. Bottom-left: rollout error growth across models. Bottom-right: Neural ODE attractor reconstruction.

TABLE I
MAIN TEST METRICS.

Model	One-step MSE	RMSE_{50}	RMSE_{100}	RMSE_{500}
MLP	3.72×10^{-4}	0.731	0.894	6.395
ResNet	5.71×10^{-5}	0.510	0.641	6.842
Neural ODE	2.18×10^{-4}	**0.438**	**0.622**	**6.011**

III. RESULTS

Table I summarizes the main quantitative results. The ResNet achieves the best one-step MSE, confirming that increment prediction is an effective local parameterization for short-horizon forecasting. However, the Neural ODE achieves the lowest rollout RMSE at all reported horizons, showing that the best local predictor is not necessarily the best long-horizon dynamical model.

Figure 3 summarizes the full story. The top-left panel shows that predicted and true values lie almost exactly on the diagonal for the strongest one-step model, illustrating how convincing short-horizon prediction can appear in a chaotic system. The top-right panel places that result in context: ResNet is strongest on one-step prediction, while the Neural ODE becomes more favorable once rollout and dynamical fidelity are considered.

The bottom-left panel makes the short-horizon versus long-horizon distinction clear. Rollout error increases for all models, but the Neural ODE accumulates error more slowly and remains more stable over time, while the MLP degrades most quickly. This highlights the gap between local prediction quality and recursive rollout behavior in learned chaotic dynamics.

The bottom-right panel adds the attractor perspective. Even when pointwise trajectories no longer match exactly, the Neural ODE still preserves the global two-lobe geometry of the Lorenz attractor. Attractor-density comparisons showed the same pattern, with ResNet and Neural ODE both clearly outperforming MLP. Robustness experiments under step-size shift and observation noise also favored the Neural ODE.

IV. DISCUSSION

This experiment illustrates a central SciML lesson. For chaotic systems, one-step error alone is not enough. A model may appear highly accurate in local forecasting while still failing to produce stable long-horizon rollouts or faithful attractor geometry. In this project, ResNet is strongest when the task is framed as short-horizon discrete prediction, while Neural ODE is strongest when the goal is long-horizon dynamical consistency and robustness under perturbation.

These results also show that different model classes capture different aspects of the system. The ResNet is highly effective as a local predictor, whereas the Neural ODE provides a better balance between rollout stability, attractor preservation, and robustness.

More broadly, the Lorenz benchmark shows why learned chaotic dynamics should be evaluated from multiple perspectives: local accuracy, rollout stability, attractor fidelity, and robustness. Exact long-horizon trajectory matching is intrinsically fragile in chaotic systems, so meaningful evaluation should emphasize both predictive accuracy and preserved dynamical structure. For chaotic dynamics, local accuracy is necessary, but not sufficient.